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Topic 1: Introduction to HCI & Interaction Design

1. Concepts to Understand

(Read this to understand the main ideas in simple English.)

- **What is HCI (Human-Computer Interaction)?**
    1. It is the study of how people (humans) use computers.
    2. It is not just about writing code. It is about making technology easy, safe, and helpful for people.
    3. It combines many subjects: **Psychology** (how we think), **Design** (how things look), and **Computer Science** (how things work).
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- **Why is HCI Important?**
    1. If a design is bad, people will be slow, make mistakes, or feel angry.
    2. If a design is good, people work faster (productivity) and feel safer.
    3. **Example:** A car with confusing buttons is dangerous. A car with clear buttons is safe.
-
- **Interaction Design (ID):**
    1. This is the "Umbrella" term (the big category).
    2. It means designing products that help people communicate and interact in their daily lives.
    3. **HCI** is just one part of Interaction Design.
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- **Good vs. Bad Design:**
    1. **Good Design:** Easy to use, obvious, usually takes only one step to do a task (Example: A simple marble answering machine).
    2. **Bad Design:** Confusing, buttons look the same, or you have to guess what to do (Example: A lift where buttons and labels look identical).
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- **The User Experience (UX):**
    1. This is about how the user **feels** when using a product.
    2. Are they happy? Frustrated? Bored?
    3. You cannot design the experience itself, but you can design *for* a good experience.
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- **Design Principles (The Rules of Design):**
    1. **Visibility:** Can the user see the control? (e.g., A hidden button is bad).
    2. **Feedback:** Does the system tell the user what happened? (e.g., A sound or light when you press a button).
    3. **Constraints:** Does the design stop you from making mistakes? (e.g., You cannot put a USB cable in the wrong way).
    4. **Consistency:** Do things look and act the same way? (e.g., "Ctrl+S" should always mean "Save").
    5. **Affordance:** Does the object's shape show you how to use it? (e.g., A handle looks like you should pull it; a button looks like you should push it).
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2. Key Points to Memorize

(Memorize these definitions, lists, and names for the exam.)

Definitions

- **HCI (Human-Computer Interaction):** A discipline concerned with the **design, evaluation, and implementation** of interactive computing systems for **human use**.
- **Interaction Design:** Designing interactive products to support the way people communicate and interact in their everyday and working lives.
- **Accessibility:** Making a product usable by people with disabilities (e.g., blind or deaf users).
- **Inclusiveness:** Making a product usable by the widest number of people possible (everyone, regardless of age or income).

Goals of HCI

1. **Safety:** (Critical for things like airplanes or hospitals).
2. **Utility:** (Does it do what is needed?).
3. **Effectiveness:** (Does it work well?).
4. **Usability:** (Is it easy to learn and use?).

The 4 Components of HCI

1. Analysis
2. Design
3. Implementation
4. Evaluation

Usability Goals (List of 6)

1. Effective to use
2. Efficient to use
3. Safe to use
4. Have good utility
5. Easy to learn
6. Easy to remember how to use

User Experience (UX) Goals

- **Desirable (Good):** Satisfying, Enjoyable, Helpful, Fun.
- **Undesirable (Bad):** Boring, Frustrating, Annoying, Confusing.

Important History & Names

- **Ivan Sutherland (1963):** Created "Sketchpad" (First Graphical User Interface).
- **Douglas Engelbart (1960s):** Invented the Computer Mouse.
- **Alan Kay (1968):** Proposed the "Dynabook" (idea for a laptop/tablet).
- **Xerox Star (1981):** First commercial computer with a GUI (windows, icons).

3. Final Summary

HCI is the study of how humans and computers interact. Its main goal is to create systems that are **safe, useful, effective, and usable**.

Interaction Design is the bigger field (umbrella) that includes HCI. It focuses on satisfying user needs and creating good **User Experiences (UX)**.

To create a good design, you must follow **Design Principles**:

1. **Visibility** (Make it visible).
2. **Feedback** (Give a reaction/sound).
3. **Constraints** (Limit errors).
4. **Consistency** (Keep it standard).
5. **Affordance** (Shape gives a clue).

Finally, designers must ensure **Accessibility** (for disabled users) and **Inclusiveness** (for everyone).

2

Topic 2: User Behaviour (Cognitive & Emotional Aspects)

1. Concepts to Understand

(Read this to understand the main ideas in simple English.)

- **What is Cognition?**
  - It simply means "how the brain works." It includes thinking, remembering, learning, and seeing.
  - Designers must understand cognition to know what users can and cannot do.
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- **Cognitive Processes:**
  - **Attention:** Focusing on one thing. If an interface is messy or has too much happening at once (multitasking), people get distracted and make mistakes.
  - **Perception:** How we see the screen. Text must be easy to read, and related items should be grouped together (using spacing or borders).
  - **Memory:** Humans have a limit on what they remember. It is easier to *recognize* something (like choosing an icon) than to *recall* it (like remembering a password).
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- **Mental Models:**
  - This is the picture people have in their heads about how a system works.
  - Sometimes these models are wrong. **Example:** People think setting an air conditioner to the lowest number cools the room *faster*, but that is wrong (it works like an on/off switch).
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- **Emotional Interaction:**
  - Design is not just about working well; it is about how it makes you *feel* (happy, trusting, or motivated).
  - If a website is beautiful, people often think it works better (this is the connection between emotion and usability).
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- **Three Levels of Design (Norman):**
  - **Visceral:** How it looks (Cool, beautiful).
  - **Behavioral:** How it works (Easy to use).
  - **Reflective:** What it means to you (Memories, self-image).
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- **Persuasive Technology:**
  - Technology designed to change your bad habits.
  - **Example:** A "digital pet" (like Pikachu) that gets angry if you don't exercise. This "nudges" you to walk more.
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2. Key Points to Memorize

(Memorize these definitions, lists, and names for the exam.)

Cognition Categories

- **Experiential vs. Reflective:** (Norman, 1993) – Doing things naturally vs. thinking hard.
- **Fast vs. Slow thinking:** (Kahneman, 2011).

Processes Definitions

- **Attention:** Selecting things to focus on from the mass of stimuli around us.
- **Perception:** How information is acquired from the world and transformed into experiences.
- **Working Memory:** Remembering a small amount of information for a short time (e.g., a conversation).
- **Long-term Memory:** Remembering things from a long time ago (e.g., a childhood song).

Recognition vs. Recall

- **Recognition:** Visual options (like menus/icons) where you just browse and select. **(Easier for users)**.
- **Recall:** Command-based interfaces where you must remember the name from memory. **(Harder for users)**.

Important Theories

- **Miller's Theory (7 +/- 2):** People can typically hold about 7 items (plus or minus 2) in their short-term memory.
    - *Note:* In UI design, you can have more than 7 items if they are grouped, because users "scan" them rather than "memorize" them.
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- **Mental Models (Craik, 1943):** Internal constructions of the world enabling predictions to be made.

Emotions & Design

- **Emotional Interaction:** Concerned with how we feel and react when interacting with technologies.
- **Affective Computing:** Computers that can recognize and express emotions (using cameras for facial expressions, sensors for sweat/pulse, or voice tone).
- **Plutchik's Wheel:** Categorizes 7 basic emotions (Anger, Disgust, Fear, Sadness, Anticipation, Joy, Surprise).
- **Persuasive Technology (Fogg):** Interactive systems designed to change people's **attitudes** and **behaviors**.

Techniques to Detect Emotion

- Facial coding (reading facial expressions).
- Biosensors (finger pulse, GSR - sweat response).
- Speech analysis (tone, pitch).
- Eye-tracking.

3. Final Summary

Topic 2 covers how the human mind works (**Cognition**) and how feelings affect computer use (**Emotion**).

In **Cognition**, we study **Attention** (don't distract the user), **Perception** (group things clearly), and **Memory** (prefer **Recognition** over Recall). Designers must respect **Mental Models**, even if users have wrong ideas about how a device works (like the thermostat/AC example).

In **Emotion**, we use **Affective Computing** to read user feelings (via cameras or sensors). Good design happens at three levels: **Visceral** (Appearance), **Behavioral** (Usability), and **Reflective** (Meaning).

Finally, computers can be used for **Behavioral Change** (Persuasive Technology) to help people stop bad habits (like using a digital pet to encourage exercise) or save energy (using visual displays of power usage).

3

Topic 3: Conceptualizing Interaction

1. Concepts to Understand

(Read this to understand the main ideas in simple English.)

- **Conceptual Model:**
  - This is a "high-level plan" or a map. It describes how a system works and is organized.
  - It helps designers think clearly *before* they start drawing buttons (widgets).
  - If the conceptual model is good, the user will find the system easy to understand.
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- **Interface Metaphors:**
  - This means using something familiar from real life to help users understand a new digital product.
  - **Example:** A "Desktop" on a computer (files, folders, trash can). It helps users know what to do because they know what a real desk looks like.
  - **Benefits:** Makes learning easier and faster.
  - **Problems:** Sometimes they break the rules (e.g., a trash can on a desktop is weird in real life). It can also limit the designer's creativity.
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- **Direct Manipulation:**
  - This means the user interacts directly with the objects on the screen.
  - **Example:** Dragging a file into a folder. You see it move.
  - It feels natural, errors are rare, and users feel in control.
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- **Social Mechanisms:**
  - **Turn-taking:** In a conversation, one person speaks, then the other. (Rule: A speaker chooses the next speaker, or someone else starts talking).
  - **Back-channeling:** Sounds like "uh-huh" or "hmm" to show you are listening.
  - **Repair:** Fixing a misunderstanding (e.g., "Huh? Did you mean X?").
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- **Presence types:**
  - **Telepresence:** Being at a party via a robot screen while you are actually at home.
  - **Social Presence:** Feeling like a person is really there with you in a virtual world (VR).
  - **Co-presence:** Being in the same physical room with other people.

2. Key Points to Memorize

(Memorize these definitions, lists, and names for the exam.)

Interaction Types (List of 5)

(These describe WHAT the user is doing)

1. **Instructing:** Telling the system what to do (e.g., commands, print file).
2. **Conversing:** Talking to the system like a human (e.g., Chatbots, Siri).
3. **Manipulating:** Dragging, opening, or zooming in on objects (e.g., Moving a file).
4. **Exploring:** Moving through a virtual or physical space (e.g., Virtual Reality or Pokemon Go).
5. **Responding:** The system starts the interaction, and you answer (e.g., An alert notification).

Interface Types (Examples of Interface Styles)

(This is HOW the interface looks or works)

1. **Command Line:** Typing text commands (fast, efficient, but hard to learn).
2. **GUI (Graphical User Interface):** WIMP (Windows, Icons, Menus, Pointer).
3. **Multimedia:** Mix of text, video, sound, graphics.
4. **Virtual Reality (VR):** Fully artificial 3D environment.
5. **Augmented Reality (AR):** Virtual objects overlaid on the real world (e.g., Pokemon Go, virtual try-ons).
6. **Mobile:** Handheld devices, issues include small screens and fat fingers.
7. **Voice:** Siri/Alexa. Good for visually impaired users.
8. **Pen-based:** Handwriting or drawing (e.g., Livescribe pen).
9. **Touch/Multi-touch:** Swiping, pinching, zooming (smartphones, tabletops).
10. **Touchless (Gesture):** Waving hands in the air (e.g., Microsoft Kinect in surgery).
11. **Haptic:** Feedback you can feel (vibration, force).
12. **Shareable:** Large screens multiple people use at once (Smartboards).
13. **Wearables:** Smartwatches, Google Glass, smart clothing.
14. **Robots:** Remote (bomb disposal), Domestic (vacuum), Social (pets).
15. **Brain-Computer Interface (BCI):** Controlling a cursor using brain waves.

Important Terms & Definitions

- **WIMP:** Stands for **W**indows, **I**cons, **M**enus, **P**ointing device.
- **Skeuomorphism:** Making digital objects look exactly like real-world objects (e.g., a digital book page turning like paper).
- **Fitts' Law:** A rule used to determine the correct size/spacing of buttons (hit area) on mobile phones (usually 44x44 points).
- **The Uncanny Valley:** When a robot looks *almost* human but not quite, it looks creepy/scary.

Menu Styles

- **Flat list:** Good for small screens (iPod).
- **Drop-down / Cascading:** Good for organizing many options.
- **Contextual:** Pop-up menu (right-click).
- **Mega Menu:** 2D drop-down showing all options at once.

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3. Final Summary

Conceptualizing Interaction involves creating a plan (**Conceptual Model**) before designing. Designers often use **Metaphors** (like a Desktop or a Shopping Cart) to help users understand the system.

There are **5 main Interaction Types**: Instructing, Conversing, Manipulating, Exploring, and Responding.

We have moved from old **Command Line** interfaces to **GUIs** (Windows & Icons), and now to **Natural User Interfaces (NUI)** like Touch, Voice, Gesture, and Brain interfaces. Future trends include **Holographic**, **Shape-Changing**, and **Smart** interfaces (AI).

Social Interaction is critical. Tech supports communication through **Telepresence** (robots), **Video Conferencing**, and **Social Games**. Designers must follow social rules (turn-taking) to avoid awkward interactions.

4

Topic 2: Interaction Design Process and Practice

1. Concepts to Understand

(Read this to understand the main ideas in simple English.)

- **Interaction Design (ID) is a Process:**
  - It is not just about making things look good. It is a cycle of finding out what is needed, designing a solution, creating a prototype, and testing it.
  - It focuses on **users** and their **goals**.
  - It involves **trade-offs**. For example, if you want a system to be very secure, it might become harder to use. You have to balance these requirements.
-
- **The Double Diamond Model:**
  - This is a famous map of the design process. It has 4 steps:
    1. **Discover:** Research the problem. Don't guess; talk to people.
    2. **Define:** clearly explain the problem using the info you found.
    3. **Develop:** Create many different ideas and solutions.
    4. **Deliver:** Test the solutions, pick the best one, and improve it.
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- **User-Centered Design:**
  - This means focusing on the user from the very beginning.
  - You don't just design once; you design, test, fix, and test again. This repeating process is called **Iterative Design**.
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- **Stakeholders:**
  - These are **not just the users**. A stakeholder is *anyone* who affects or is affected by the project.
  - **Examples:** Users, the boss, the developers, the people making the laws, etc.
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- **AgileUX:**
  - Traditional software development happens in stages (step 1, then step 2).
  - **Agile** development happens in short cycles called **Sprints** (usually 1–3 weeks).
  - **AgileUX** puts User Experience (UX) research into these fast sprints.
  - **Dual Track:** This is a technique where the Designers work one step ahead of the Programmers (Developers). Designers plan the next sprint while developers build the current one.
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- **Design Patterns:**
  - These are solutions to common problems that already exist. You don't need to invent a new way to turn a page; you just use "pagination" (page numbers) or "infinite scroll."
  - **Dark Patterns:** These are "tricks" used to fool users (e.g., making it very hard to unsubscribe from an email).
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2. Key Points to Memorize

(Memorize these definitions, lists, and names for the exam.)

4 Basic Activities of Interaction Design

1. **Discovering requirements** (What do people need?).
2. **Designing alternatives** (Creating different ideas).
3. **Prototyping** (Building rough versions to test).
4. **Evaluating** (Testing the product with users).

3 Principles of People-Centered Design

1. **Early focus on users and tasks.**
2. **Empirical measurement** (Testing with real users to get data).
3. **Iterative design** (Design -> Test -> Fix -> Repeat).

4 Approaches to Design

1. User-centered design.
2. Activity-centered design.
3. Systems design.
4. Genius design.

Stakeholder Types

- **Direct users:** The people actually pressing the buttons.
- **Indirect users:** People affected by the system but who don't touch it.
- **Others:** Developers, executives, legislators.

Agile Terms

- **Sprint:** A short timebox (1-3 weeks) to finish a specific task.
- **Dual Track:** Designers work one iteration *ahead* of developers.
- **Technical Debt (in UX):** Shortcuts taken now that will cause problems later (e.g., skipping research to be fast).

Design Pattern Types

- **Pattern:** A good solution to a known problem.
- **Anti-pattern:** A bad solution; don't do this (e.g., using "Click Here" instead of descriptive text).
- **Dark Pattern:** A deliberate trick to manipulate the user.

Design Tools (Examples)

- Sketch, Balsamiq, Axure, Figma.

3. Final Summary

Topic 2 focuses on the **Process** of design. The main goal is **User-Centered Design**, which involves an **Iterative** cycle (design, test, repeat) described by the **Double Diamond** model (Discover, Define, Develop, Deliver).

To be successful, you must identify all **Stakeholders**, not just the users. In modern software companies, teams use **AgileUX**, where they work in short **Sprints** using a **Dual Track** system (Design is one step ahead of Code).

Designers save time by using **Design Patterns** (standard solutions) but must avoid **Dark Patterns** (tricks). They use tools like **Figma** or **Sketch** to create prototypes for testing.

5

Topic 5: Data Gathering & Discovering Requirements

1. Concepts to Understand

(Read this to understand the main ideas in simple English.)

- **Why Gather Data?**
  - Designers cannot just guess what users want. They must collect real information.
  - You must set clear **goals** (what you want to know) and choose the right **participants** (who to ask).
- **Triangulation:**
  - This means not trusting just *one* method. You should use different methods (e.g., interviews + observation) or different researchers to get the true picture. It makes data better.
- **Pilot Studies:**
  - This is a "practice run." Before doing a big survey or interview, you test it with a small group to check for mistakes.
- **Interview Styles:**
  - **Unstructured:** Like a casual chat. You discover new things, but it is hard to compare results.
  - **Structured:** Following a strict script (like a spoken questionnaire).
  - **Semi-structured:** You have a script, but you can ask extra questions if something interesting comes up. This is usually the best balance.
- **Direct vs. Indirect Observation:**
  - **Direct:** You watch the user with your own eyes (either in the "wild" real world or a lab).
  - **Indirect:** You use technology to watch them (tracking logs, video cameras, or diaries they write).
- **Requirements:**
  - A "requirement" is a statement about what the product *must do* or how it must perform.
  - It stops the team from misunderstanding each other.
- **Functional vs. Non-Functional Requirements:**
  - **Functional:** What the system *does* (e.g., "The button prints the page").
  - **Non-functional (Characteristics):** *How* it does it (e.g., "It must be safe," "It must hold 100 photos").
- **Personas:**
  - A fake person created based on real data. It represents a "typical user."
  - It helps designers talk about who they are building for (e.g., "Would Persona Bob like this?").
- **Scenarios:**
  - A story about how the Persona uses the product. It helps explore how the system works in real life.

2. Key Points to Memorize

(Memorize these definitions, lists, and rules for the exam.)

6 Key Issues in Data Gathering

1. Setting goals (What do we want to know?).
2. Identifying participants (Who do we ask?).
3. Relationship with participants (Informed consent, professional behavior).
4. Ethical considerations (Privacy, data storage).
5. Triangulation (Using multiple views).
6. Pilot studies (Testing the study first).

Ways to Capture Data

- Notes
- Audio recording
- Video recording
- Photographs

Question Types (for Interviews & Questionnaires)

- **Closed Questions:** Have fixed answers (Yes/No, Multiple choice). *Easier to analyze.*
- **Open Questions:** The user can write whatever they want. *Harder to analyze but gives more detail.*
- **Likert Scale:** A range of agreement (e.g., 1 to 5, Strongly Disagree to Strongly Agree).

Frameworks for Observation

- **Simple Framework:** The Person, The Place, The Thing.
- **Contextual Inquiry:** The researcher acts like an **Apprentice**, and the user acts like a **Master**. You watch and learn from them as they work.

The 4 Types of Requirements

1. **Functional:** What the system must do.
2. **Data:** What info needs to be stored (e.g., names, numbers).
3. **Environment (Context):**
    - *Physical* (is it dusty/noisy?)
    - *Social* (do people share data?)
    - *Technical* (what hardware does it run on?)
4.
5. **User Characteristics:** Who is the user? (Novice vs. Expert).

User Stories Format

Used in agile development to describe a requirement:

- "As a **[role]**, I want **[behavior]** so that **[benefit]**."

Volere Shell

- A template or card used to write down specific requirements systematically.

Use Cases

- A step-by-step list of interactions.
- Shows the **User Intention** (what the user does) vs. the **System Responsibility** (what the system does).

Brainstorming Rules (Alex Osborn)

1. Quantity over quality (Get many ideas).
2. Criticisms withheld (Don't say an idea is bad).
3. Encourage "wild" ideas (Think outside the box).
4. Combine and refine ideas.

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3. Final Summary

Data Gathering is the first step to understand users. We must set goals, be ethical, and run **Pilot Studies**. We use **Triangulation** (using multiple methods) to ensure data is correct.

The main methods are:

- **Interviews** (Asking people).
- **Questionnaires** (Surveys with Likert scales).
- **Observation** (Watching people directly or indirectly using logs/diaries).
- **Ethnography** (Living with the users to understand their culture).

Once data is gathered, we find the **Requirements**. These tell us **what** the system should do (**Functional**) and **how** it should operate (**Environment, Data, User Profile**).

To make these requirements easy to understand, designers use:

- **Personas** (Fake user profiles).
- **Scenarios** (Stories of usage).
- **Use Cases** (Step-by-step lists of actions).
- **User Stories** (Short descriptions of a goal).

6

Topic 6: Persona, Scenario & Task Analysis

1. Concepts to Understand

(Read this to understand the main ideas in simple English.)

- **Personas (The "Who"):**
  - Designers cannot design for "everyone." If you design for everyone, you design for no one.
  - A **Persona** is a fake (fictional) character that represents a real group of users.
  - It helps the team focus. Instead of saying "Users want this," you say "Bob wants this."
  - You create them using real data (interviews/research), not by guessing.
-
- **Empathy Map:**
  - A visual tool to help understand the Persona better. It looks at what the user **Sees**, **Hears**, **Thinks**, **Says**, and what creates **Pain** or **Gain** for them.
-
- **Task Analysis (The "What"):**
  - This is the study of how people do their jobs or achieve their goals.
  - It breaks a user's activity down into small pieces to understand it completely.
  - It helps researchers analyze **existing** situations.
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- **Hierarchical Task Analysis (HTA):**
  - "Hierarchical" means organized in levels (Top to Bottom).
  - HTA involves breaking a big **Goal** (e.g., Withdrawing cash) into smaller **Tasks**, and then into tiny **Actions**.
  - It uses a diagram (boxes and lines) or a text list.
-
- **Plans in HTA:**
  - A list of subtasks is not enough. You need to know the **order**.
  - A **Plan** tells you the rule for doing the steps.
  - *Example:* "Do step 1, then step 2. If required, do step 3."
-
- **Scenarios (The "Story"):**
  - A Scenario is a story. It describes a user (Persona) completing a task in a specific situation.
  - It is written in simple, narrative language (like a book), not technical code.
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2. Key Points to Memorize

(Memorize these definitions, lists, and rules for the exam.)

Persona Facts

- **Definition:** A fictional character representing a major user group.
- **Benefits:**
    1. Simplifies decision-making.
    2. Prevents assumptions (guessing) about users.
    3. Resolves design disagreements.
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- **Limitations:** They are **not reusable**. A persona made for one product cannot be used for a different product.
- **How to create:** Use background data, use empathy maps, add a real photo, and give them a name.

Empathy Map Sections

1. **Think & Feel:** What matters to them?
2. **See:** What is in their environment?
3. **Hear:** What do friends/bosses say?
4. **Say & Do:** How do they behave?
5. **Pain:** Fears and obstacles.
6. **Gain:** Wants and measures of success.

Task Analysis vs. Task Description

- **Task Analysis:** Investigates an **existing** situation (what people do now).
- **Task Description:** Used to envision a **new** system.

Hierarchical Task Analysis (HTA)

- **Structure:**
    - **Goal:** The state the human wants to achieve (e.g., Buy a DVD).
    - **Task:** The structured set of activities to do it.
    - **Action:** The simple steps (e.g., click button, insert card).
-
- **Plans:** Describe *how* to do the tasks. Types of plans include:
    - **Sequence:** 1.1 then 1.2 then 1.3.
    - **Selection (Decision):** Do 1 or 2.
    - **Cycles:** Repeat until finished.
    - **Parallel:** Do tasks at the same time.
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Scenarios

- **Definition:** A narrative description describing how a user intends to use a system.
- **Writing Process:**
    1. Determine target group.
    2. List tasks.
    3. Rewrite tasks into a **simple story**.
    4. **Important Rule:** Omit (remove) links to technology. Focus on the behavior, not the technical buttons.
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3. Final Summary

Topic 6 connects **Personas** (the user), **Task Analysis** (the breakdown of work), and **Scenarios** (the story of use).

1. **Personas** define **WHO** we are designing for. We use tools like **Empathy Maps** to understand their needs and pains.
2. **Task Analysis (HTA)** defines **WHAT** needs to be done. We break a main **Goal** down into **Tasks** and **Sub-tasks**. We use **Plans** to show the order of these steps (sequence, repetition, etc.).
3. **Scenarios** describe **HOW** the interaction happens in a story format. They combine the Persona and the Task into a narrative without using technical words.

7

Topic 7: Design, Visual Perception & Principles

1. Concepts to Understand

(Read this to understand the main ideas in simple English.)

- **Interface Metaphors:**
  - This is when a digital design looks or acts like a real-world object to help users understand it.
  - **Example:** A "Card" looks like a physical business card. It is easy to sort, flip, and organize.
  - **Example:** A "Shopping Cart" helps users understand they are collecting items to buy.
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- **Information Architecture (IA):**
  - This is the science of organizing and labeling content so users can find what they need.
  - **Card Sorting:** A method where you ask users to group topics together. This helps designers know how to categorize menus or website sections.
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- **Visual Perception (Gestalt Principles):**
  - This explains how humans see and organize visual information. Our brains try to find patterns to make sense of what we see.
  - **Similarity:** Things that look the same (shape, color) are seen as a group. *Note: Color is the strongest way to show relationship.*
  - **Proximity:** Things that are **close together** are seen as related. *Note: Proximity is stronger than similarity.*
  - **Enclosure:** Putting a border or box around items tells the user they belong together.
  - **Continuation:** Our eyes naturally follow lines or curves (like breadcrumbs on a website).
  - **Figure-Ground:** We separate the main object (Figure) from the background (Ground). Shadowing and highlighting help with this.
  - **Law of Prägnanz (Simplicity):** Humans prefer simple shapes. We interpret complex images in the simplest form possible.
-
- **Slips vs. Mistakes (Types of Errors):**
  - **Slip:** You have the **right goal** but do the **wrong action** because you were on "autopilot" (e.g., typing 'i' instead of 'o', or putting soap on your toothbrush instead of toothpaste).
  - **Mistake:** You have the **wrong goal**. You understood the system incorrectly. (e.g., thinking a light means "low oil" when it actually means "low tire pressure").
-

2. Key Points to Memorize

(Memorize these definitions, lists, and facts for the exam.)

Components of Information Architecture (IA)

1. **Organisation Schemes:** How you categorize info.
2. **Labeling Systems:** What you call the info.
3. **Navigation Systems:** How users move through the app/site.
4. **Search Systems:** How users look for specific info.

UI Element Categories

- **Input Controls:** Buttons, checkboxes, radio buttons, text fields.
- **Navigational:** Breadcrumbs, sliders, search fields, pagination, tags.
- **Informational:** Tooltips, icons, notifications, progress bars.
- **Containers:** Accordions (expandable lists).

Ways to Prevent Errors

- **Good Defaults:** Pre-fill commonly used info so users type less.
- **Forgiving Formatting:** Let users type data their way (e.g., phone numbers with or without spaces).
- **Constraints:** Block invalid actions (e.g., grey out dates you cannot book).
- **Offer Suggestions:** "Did you mean...?"
- **Confirm Destructive Actions:** Ask "Are you sure?" before deleting.
- **Warn Before Errors:** Show a limit (e.g., "10 characters left").

Design Guidelines (Industry Standards)

- **Shneiderman’s 8 Golden Rules:** Includes rules like "Strive for consistency" and "Offer informative feedback."
- **Norman’s 7 Principles:** Includes "Make things visible" and "Get the mappings right."
- **Google Material Design:** Uses the concept of "Material as Metaphor" (tactile reality, paper/ink physics, shadows).
- **Apple Human Interface Guidelines:** Focuses on "Clarity," "Deference" (content is king), and "Depth."

3. Final Summary

Topic 7 focuses on how to **Design** interfaces that humans can easily understand.

First, we organize information using **Information Architecture** and techniques like **Card Sorting**. We use **Metaphors** (like Cards or Shopping Carts) to make digital objects feel familiar.

Second, we apply **Gestalt Principles** (Proximity, Similarity, Enclosure, Continuation, Figure-Ground) to make layouts visually clear. For example, items placed close together (**Proximity**) are seen as a group.

Third, we design to prevent user **Errors**. Errors are either **Slips** (accidental typos) or **Mistakes** (wrong understanding). We prevent them by using **Constraints** (blocking bad choices), **Defaults** (smart pre-filling), and **Confirmation** dialogs.

Finally, industry standards like **Google Material Design** and **Apple Guidelines** help designers create consistent and high-quality apps.

8

Topic 8: Prototyping and Construction

1. Concepts to Understand

(Read this to understand the main ideas in simple English.)

- **What is a Prototype?**
  - A prototype is a simple model of your idea. It lets users see and touch your idea before you build the final product.
  - **Goal:** To test ideas, get feedback, and save money (fixing a prototype is cheaper than fixing a finished app).
-
- **Design with People:**
  - Don't just design *for* users; design *with* them (**Co-design**).
  - **Participatory Design:** Users are partners in the design team, not just testers.
-
- **Types of Prototypes:**
  - **Low-Fidelity (Lo-Fi):**
    - Made of paper, cardboard, or sticky notes.
    - Quick, cheap, and disposable. Good for testing *basic concepts*.
    - Examples: Sketches, Storyboards, Wizard-of-Oz.
  -
  - **High-Fidelity (Hi-Fi):**
    - Made with software code or high-quality materials.
    - Looks and feels like the final product.
    - Good for testing *usability details* but takes longer to make.
  -
  - **3D Printing:** Used for making physical shapes (like a smart dress or custom shoes).
-
- **Conceptual vs. Concrete Design:**
  - **Conceptual Design:** The "Big Picture." (What will it do? What metaphors will we use? What interaction style?).
  - **Concrete Design:** The "Details." (What color is the button? Where is the menu located? What icon do we use?).
-
- **Storyboards:**
  - Like a comic strip. It shows a *sequence* of user actions to tell a story about how the product is used.
-
- **Wizard-of-Oz Prototype:**
  - A fake computer system. The user thinks they are interacting with an AI or computer, but a *human* is secretly controlling the responses from another room.
-

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2. Key Points to Memorize

(Memorize these definitions, lists, and examples for the exam.)

Compromises in Prototyping

Since a prototype is not the final product, it must "compromise" (leave things out):

1. **Horizontal:** Covers **many features** but with **little detail** (shallow). Good for showing the whole scope.
2. **Vertical:** Covers **only a few features** but with **full detail** (deep). Good for testing one specific complex task.

Construction Tools (SDKs)

- **SDK (Software Development Kit):** A box of tools for programmers to build apps for a specific platform (e.g., iOS SDK for iPhones).
- **Examples of Physical Kits:**
    - **Arduino:** An electronics board for building digital devices.
    - **LilyPad Arduino:** Washable electronics for clothes (e-textiles).
    - **MaKey MaKey:** Turns everyday objects (like bananas or play-doh) into touchpads.
    - **BBC micro:bit:** A tiny computer for teaching coding.
-

Prototyping Methods List

1. **Sketching:** Hand-drawing screens (e.g., flowcharts).
2. **Storyboarding:** Comic strip style series of scenes.
3. **Card-based:** Using index cards to represent screens. You "swap" cards when the user clicks a button.
4. **Wizard-of-Oz:** Human faking the computer response.

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3. Final Summary

Topic 8 is about building and testing ideas. Instead of building the full product immediately, we build a **Prototype**.

We start with **Low-Fidelity** prototypes (paper, sketches) to test **concepts**. Then we move to **High-Fidelity** prototypes (software) to test **details**.

We use methods like **Storyboarding** to visualize user scenarios and **Wizard-of-Oz** to test advanced tech without coding it yet. When moving to the final **Construction** phase, developers use **SDKs** (Software Development Kits) and hardware kits like **Arduino** or **micro:bit**.

Designing involves two levels: **Conceptual Design** (the idea/metaphor) and **Concrete Design** (the actual colors/buttons). We must compromise: we can build wide-and-shallow (**Horizontal**) or narrow-and-deep (**Vertical**) prototypes to suit our testing needs.

9

Topic 9: Evaluation Methods

1. Concepts to Understand

(Read this to understand the main ideas in simple English.)

- **Why Evaluate?**
  - Designers often *guess* that their design is good. Evaluation *proves* it.
  - It checks if the product is usable, effective, and gives a good user experience.
-
- **Controlled vs. Natural Settings:**
  - **Controlled:** A laboratory environment. The user is given specific tasks, and the environment is artificial (no distractions). Good for measuring performance (time, errors).
  - **Natural (In-the-Wild):** Real-world usage (e.g., at home, on a bus). Good for understanding behavior and context, but hard to control.
  - **Remote:** Testing from a distance (using Zoom, online tools). Essential during pandemics or for reaching international users.
-
- **Usability Testing vs. Research Experiments:**
  - **Usability Testing:** Finds and fixes problems to improve a *product*. Typically uses few participants (5-10).
  - **Research Experiments:** Tests a hypothesis to discover new *knowledge*. Requires many participants and strict statistical validation.
-
- **Heuristic Evaluation (Inspection):**
  - This is when *experts* check the design against a set of rules (Heuristics). It does NOT involve real users.
  - Experts pretend to be users and look for violations of design principles.
-
- **Cognitive Walkthrough:**
  - Experts walk through a task step-by-step to see if a new user would know what to do.
  - They ask 3 key questions at each step:
    1. Will the user know *what* to do?
    2. Will the user *see* how to do it?
    3. Will the user understand the *response*?
  -
-
- **Analytics & A/B Testing:**
  - **Analytics:** Using software (like Google Analytics) to track what thousands of users do (clicks, time on page).
  - **A/B Testing:** Showing two different versions of a design (Version A and Version B) to thousands of users to see which one performs better.
-
- **Predictive Models (Fitts' Law):**
  - A mathematical formula used to predict how fast a user can click a button.
  - Bigger and closer targets are faster to hit.

2. Key Points to Memorize

(Memorize these definitions, lists, and rules for the exam.)

Types of Evaluation

1. **Controlled settings:** Usability labs & experiments (Direct user involvement).
2. **Natural settings:** Field studies & in-the-wild (Direct user involvement).
3. **Without users:** Inspections, models, analytics (Indirect/Expert involvement).

Evaluation Methods Table

- **Observing:** Can be done in controlled, natural, or without users.
- **Asking users:** Controlled (interviews) or natural settings.
- **Asking experts:** Without users (inspections).
- **Testing:** Controlled settings.
- **Modeling:** Without users (predictive).

Key Metrics (What to Measure)

- Time to complete a task.
- Number of errors.
- Success rate.
- User satisfaction ratings.

Evaluation Heuristics (Nielsen)

1. **Visibility of system status.**
2. **Match between system and real world.**
3. **User control and freedom.**
4. **Consistency and standards.**
5. **Error prevention.**
6. **Recognition rather than recall.**
7. **Flexibility and efficiency of use.**
8. **Aesthetic and minimalist design.**
9. **Help users recognize, diagnose, and recover from errors.**
10. **Help and documentation.**

Nielsen's "Why 5 Users?" Rule

- Testing with just **5 users** finds about **85%** of usability problems.
- After 5 users, you start seeing the same problems repeatedly (diminishing returns).

Fitts' Law Rule

- **Formula logic:** Time to click depends on **Distance** (how far away it is) and **Size** (how big it is).
- **Design Rule:** Make important buttons **large** and placing them **close** to the starting cursor position improves speed.

Living Labs

- Places (homes/cities) equipped with sensors to study technology use in everyday life over long periods.

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3. Final Summary

Evaluation is how we check if a design works. It can happen in **controlled labs** (for precision) or **natural settings** (for realism).

We use **Usability Testing** with small groups (5 users) to fix products, and **Research Experiments** with large groups to test theories.

Sometimes we evaluate *without* users. Experts can do a **Heuristic Evaluation** using Jacob Nielsen's 10 Heuristics (like "Error Prevention" or "Consistency"). They can also use **Cognitive Walkthroughs** to test ease of learning.

We also use data. **Analytics** tracks user clicks, and **A/B Testing** compares two designs at scale. Finally, laws like **Fitts' Law** allow us to predict performance mathematically (e.g., bigger buttons are faster to click).

10

Topic 10: Data Analysis, Interpretation & Presentation

1. Concepts to Understand

(Read this to understand the main ideas in simple English.)

- **Quantitative vs. Qualitative Data:**
    - **Quantitative:** Deals with **Numbers**. (e.g., "User took 10 seconds," "80% of users clicked yes"). You use math to analyze this.
    - **Qualitative:** Deals with **Words and meanings**. (e.g., "The user looked confused," "The user said they felt happy"). You look for themes and patterns to analyze this.
-
- **Quantitative Analysis (Math for HCI):**
    - You often calculate averages to understand performance.
    - **Mean:** The mathematical average. (Add all up and divide).
    - **Median:** The middle number. (Good if there are extreme outliers, like one person taking 2 hours while everyone else took 5 minutes).
    - **Mode:** The most common number.
-
- **Qualitative Analysis (Finding Meaning):**
    - **Coding:** This does *not* mean computer programming. It means reading interview notes and assigning a label (code) to interesting parts (e.g., labeling a complaint as "Network Error").
    - **Themes:** Grouping codes together to find the big story (e.g., "All users struggle with the login page").
-
- **Analytical Frameworks (Different ways to look at data):**
    - **Grounded Theory:** You start with the data and let the theory grow from it. You don't start with a hypothesis.
    - **Distributed Cognition:** This theory says that "thinking" doesn't just happen inside your head. It involves other people, papers, and tools you are using. (e.g., A pilot flying a plane thinks using the cockpit dials and the co-pilot).
    - **Conversation Analysis:** Looking very closely at how people talk (pauses, interruptions, "umms" and "ahhs").
-
- **Presenting Findings:**
    - Once you analyze the data, you must tell a story. Don't just show a big table of numbers. Use charts, simple summaries, and stories to make stakeholders understand.
-

2. Key Points to Memorize

(Memorize these definitions, lists, and classifications for the exam.)

Types of Data

- **Quantitative:** Expressed as numbers.
- **Qualitative:** Expressed as words/images/themes.

The 3 Types of Averages

1. **Mean:** Mathematical average.
2. **Median:** Middle value.
3. **Mode:** Most frequently occurring value.

Qualitative Analysis Frameworks

- **Content Analysis:** Counting how often a specific word or image appears (Categorizing).
- **Discourse Analysis:** Looking at the meaning of what is said (Language as a tool).
- **Conversation Analysis:** Examining the tiny details of dialogue (semantics).
- **Interaction Analysis:** Studying empirical data (videos) collaboratively in a team.

Grounded Theory Steps

(A systematic way to analyze qualitative data)

1. **Open Coding:** Identifying categories.
2. **Axial Coding:** Linking categories together.
3. **Selective Coding:** Forming a final theoretical scheme (the main story).

System-based Frameworks

- **Socio-technical Systems:** Understanding the technical system + the people/social aspect together.
- **Distributed Cognition:** Information is processed across individuals and artifacts (tools/screens).

Tools to Support Analysis

- **SPSS / R:** Used for Statistical (Quantitative) analysis.
- **Nvivo:** Used for Qualitative (text) analysis.
- **Web Analytics:** (Google Analytics) Visualizes user behavior on websites (Time spent, clicks).

3. Final Summary

Topic 10 explains what to do with the data you gathered in Topic 9.

If you have **Quantitative Data** (numbers), you use statistics (Mean, Median, Mode) and charts to find patterns or outliers (weird results). Fitts' Law and Web Analytics are common tools here.

If you have **Qualitative Data** (words), you use **Coding** to find themes. Frameworks like **Grounded Theory** help structure this process (Open -> Axial -> Selective).

For complex systems involving teams and tools, frameworks like **Distributed Cognition** help analyze how people, technology, and the environment work together.

Finally, you must **Present** your findings clearly. Use visual charts for numbers and tell stories for qualitative insights to ensure the design team understands what to fix.

Exam Preparation

THE HCI & INTERACTION DESIGN MASTER DOCUMENT

Semester 1, 25/26 Exam Preparation

🛑 SECTION 1: THE FUNDAMENTALS (Topics 1 & 7)

These concepts usually appear in Multiple Choice Questions (MCQ) or Part A Short Answers.

1. What is HCI?

Definition: Human-Computer Interaction (HCI) is the discipline concerned with the **design, evaluation, and implementation** of interactive computing systems for **human use**.

- **Why is it important?** Because computers are now used by everyone, not just experts. Poor design leads to accidents (safety), wasted time (inefficiency), and frustration.

2. Interaction Design (ID)

ID is the "umbrella" term. It is broader than HCI. It means designing products to support the way people communicate and interact in their daily lives.

- **The Goal:** To create **good User Experiences (UX)**.

3. The Two Types of Goals (MEMORIZE THESE LISTS)

In the exam, you will likely be asked to "List 5 Usability Goals" or "Distinguish between Usability and UX goals."

A. Usability Goals (The Practical Stuff)
Use the acronym "E-E-S-E-E"

1. **Effective:** Does it do what it is supposed to do?
2. **Efficient:** Can users do it quickly (few steps)?
3. **Safe:** Does it prevent errors or recover from them?
4. **Easy to learn:** Can a beginner use it immediately?
5. **Easy to remember:** Can I return after a week and remember how to use it?

B. User Experience (UX) Goals (The Emotional Stuff)
These describe how the user **feels**.

- **Positive:** Satisfying, Enjoyable, Engaging, Helpful, Motivating, Fun.
- **Negative:** Boring, Frustrating, Annoying, Confusing.

4. Design Principles (The Golden Rules)

Exam Tip: You may see a picture of a bad interface and be asked "Which principle is violated?"

1. **Visibility:** Can the user see the control? (e.g., A hidden button violates Visibility).
2. **Feedback:** Sending information back to the user about what has been done. (e.g., A button making a 'click' sound, or a loading bar).
3. **Constraints:** Restricting possible actions to prevent errors. (e.g., You cannot select a date in the past on a calendar).
4. **Consistency:** Interfaces should look and act the same. (e.g., 'Ctrl+S' should always mean Save, buttons should always be in the same place).
5. **Affordance:** The **shape** of an object tells you how to use it.
    - A button *looks* like it should be pushed.
    - A handle *looks* like it should be pulled.
    - **Anti-pattern:** A door plate that looks like a handle (User pulls it, but it doesn't open). This is **Bad Affordance**.
6.

🧠 SECTION 2: THE HUMAN USER (Topics 2 & 3)

Understanding the brain helps us design better interfaces.

1. Cognition (Thinking)

Designers must understand user limitations.

- **Attention:** Users cannot focus on everything. **Don't clutter the screen.**
- **Perception:** How users see things. Group related items together.
- **Mental Models:** The picture the user has in their head about how a system works.
    - *The Thermostat Mistake:* People think turning a heater to "Max" heats the room *faster*. It doesn't; it just heats it *longer*. The user has the wrong mental model. Good design corrects this.
-

2. Memory: Recognition vs. Recall (CRITICAL)

- **Recall:** You must retrieve information from your brain without help. (e.g., Command Line Interface: You must remember ls -a to see files). **This is HARD.**
- **Recognition:** You see the answer and pick it. (e.g., Graphical User Interface: You see a file icon and click it). **This is EASY.**
- *Design Rule:* Always prefer Recognition over Recall.

3. Interaction Types

How does the user talk to the system?

1. **Instructing:** Typing commands (Command Line). Fast for experts, hard for novices.
2. **Conversing:** Using natural language (Chatbots, Siri). Good for novices, but can be ambiguous.
3. **Manipulating:** Acting on objects (WIMP - Windows, Icons, Menus, Pointers). Dragging files into a trash can. Good for visualization.
4. **Exploring:** Moving through a 3D environment (VR/Games).

🎨 SECTION 3: VISUAL DESIGN & INTERFACE STYLES (Topic 7)

1. Gestalt Principles (Visual Perception)

How do humans organize what they see?

- **Proximity:** Objects that are close together are seen as a group.
- **Similarity:** Objects that look the same (color, shape) are seen as a group.
- **Closure/Enclosure:** We see objects inside a box/border as a group.
- **Figure-Ground:** We separate the main object (Figure) from the background (Ground).
- **Continuation:** Our eyes follow lines and curves (like a road or breadcrumbs).

2. Interface Metaphors

Using a known concept to explain a new one.

- **Desktop:** We call the computer screen a "Desktop" with "Folders" and "Trash cans" so office workers understand it.
- **Shopping Cart:** In online stores, we "add to cart."
- *Risk:* Metaphors can constrain design. (e.g., A physical folder can't hold a movie, but a computer folder can).

3. Errors: Slips vs. Mistakes

- **Slip:** You have the **Right Goal** but do the **Wrong Action**.
    - *Cause:* You were on "autopilot" or not paying attention.
    - *Example:* You wanted to type "The", but your finger hit "Teh".
-
- **Mistake:** You have the **Wrong Goal**.
    - *Cause:* You misunderstood the system.
    - *Example:* You clicked "Format Disk" because you thought it meant "Organize Files," but it actually meant "Delete Everything."
-

🔄 SECTION 4: THE DESIGN PROCESS (Topics 5 & 6)

This is the most important section for the "Long Structured Questions." You will be given a scenario and asked to do the following.

1. The Lifecycle (Iterative Design)

Design is not a straight line; it is a circle.

1. **Discover:** Gather requirements.
2. **Design:** Create alternatives.
3. **Prototype:** Build rough versions.
4. **Evaluate:** Test with users.
   (If the evaluation fails, go back to Design).

2. Data Gathering Methods

How do we find out what users want?

- **Interviews:** Talking to users. Good for deep details.
- **Questionnaires (Surveys):** Good for reaching many people. Uses **Likert Scales** (1-5 agreement).
- **Observation:** Watching people work.
    - *Direct:* You stand there with a clipboard.
    - *Indirect:* You track their logs or use a camera.
    - *Ethnography:* You live with the users to understand their culture.
-
- **Triangulation:** Using **more than one method** (e.g., Survey + Interview) to double-check that your data is accurate.

3. Personas

A **Persona** is a fictional character based on your research. It prevents the team from designing for "everyone."

- **Example for an Exam:** If the question is "Design an ATM," your Persona might be:
    - **Name:** Mak Cik Kiah.
    - **Age:** 65.
    - **Goal:** Wants to withdraw cash for the market.
    - **Frustration:** Hates small buttons and cannot see well without glasses.

4. Scenarios

A **Scenario** is a simple story about the Persona using the product.

- *Rule:* **Do not use technical terms.** Do not say "User clicks button A."
- *Write:* "Mak Cik Kiah walks to the ATM. She is in a hurry. She inserts her card..."

5. HTA (Hierarchical Task Analysis) - *EXAM GOLD MINE*

You must learn to draw this. It breaks a goal into sub-tasks.

Scenario: Buying a Drink from a Vending Machine.

- **0. Goal:** Buy Drink.
- **1. Choose Drink.**
    - 1.1 Locate code.
    - 1.2 Enter code.
-
- **2. Pay.**
    - 2.1 Insert coins.
-
- **3. Get Drink.**
    - 3.1 Take can.
    - 3.2 Take change.
-

Plan 0: Do 1, then 2. If money sufficient, do 3.

🛠️ SECTION 5: PROTOTYPING (Topic 8)

1. What is a Prototype?

A representation of a design idea. It is not the final product.

2. Low-Fidelity vs. High-Fidelity (Compare & Contrast)

Feature	Low-Fidelity (Paper, Sketch)	High-Fidelity (Software, App)
Cost	Cheap	Expensive
Speed	Fast to build	Slow to build
Purpose	Testing concepts/layout	Testing details/usability
Disadvantage	Functionality is limited	User might think it's "finished"

3. Other Methods

- **Storyboarding:** Drawing comic strips of the user using the product (Used with Low-Fi).
- **Wizard-of-Oz:** The user thinks they are interacting with a computer, but a human is behind the scenes typing the responses. (Used to test AI or advanced systems before coding them).

4. Horizontal vs. Vertical Compromise

Prototypes cannot do everything.

- **Horizontal:** Shows **all features** (the whole menu), but they don't actually work (Shallow).
- **Vertical:** Shows **one feature** in perfect detail, but nothing else works (Deep).

🧪 SECTION 6: EVALUATION (Topic 9)

Checking if the system is good.

1. Usability Testing (Controlled)

- **Where:** In a laboratory.
- **Who:** Real users (5 to 10 is enough to find 80% of problems).
- **Data:** Quantitative (Time taken, Number of errors).
- **Pros:** Good for comparing products.
- **Cons:** Not natural. People behave differently in a lab.

2. Field Studies (Natural/In-the-Wild)

- **Where:** In the real world (e.g., Home, Bus station).
- **Who:** Real users.
- **Data:** Qualitative (Observation, Video).
- **Pros:** Shows real behavior and interruptions.
- **Cons:** Messy, hard to control noise/distractions.

3. Inspections (NO Users)

- **Heuristic Evaluation: Experts** check the interface against a list of rules (Heuristics).
- **Cognitive Walkthrough:** Experts step through a task asking "Will the user know what to do here?"

4. Nielsen’s 10 Heuristics (Memorize at least 5 for the Exam)

1. **Visibility of System Status:** (e.g., Battery icon).
2. **Match between System and Real World:** (e.g., Speak the user's language).
3. **User Control and Freedom:** (e.g., "Undo" and "Cancel" buttons).
4. **Consistency and Standards:** (e.g., Links are always blue).
5. **Error Prevention:** (e.g., Confirmation message before delete).
6. **Recognition rather than Recall:** (e.g., Menu lists).
7. **Flexibility and Efficiency:** (e.g., Keyboard shortcuts for experts).
8. **Aesthetic and Minimalist Design:** (e.g., No clutter).
9. **Help Users with Errors:** (e.g., Error messages in plain English, not "Error 404").
10. **Help and Documentation:** (e.g., Searchable FAQ).

📊 SECTION 7: ANALYSIS & MODELS (Topics 9 & 10)

1. Fitts' Law

- **The Formula:** $T = a + b \log_2 (D/W + 1)$
- **The Meaning:** The time to acquire a target depends on **Distance (D)** and **Width (W)**.
- **The Rule:** To make something faster to click, make it **Bigger (Increase W)** or **Closer (Decrease D)**.

2. Types of Data Analysis

- **Quantitative:** Numbers (Averages/Means).
    - *Mean:* Average.
    - *Median:* Middle point.
    - *Mode:* Most common.
-
- **Qualitative:** Words/Patterns.
    - *Coding:* Labeling data to find patterns.
    - *Themes:* Grouping patterns together.

📝 SECTION 8: EXAM ANSWERING STRATEGY (HOW TO PASS)

The exam will likely follow this structure. Here is how you tackle it on February 5th.

Part A: Multiple Choice & Short Answer

- **Strategy:** Speed. Don't think too hard. If you memorized the "Lists" in this document (5 Usability goals, 4 Interaction types, 10 Heuristics), you will ace this.

Part B: The Scenario Questions (The Danger Zone)

You will be given a text like: *"Siti is designing an App for elderly people to buy groceries."*

- **Question 1: "Identify the Users/Stakeholders"**
  - **Answer:** Do not just say "Old people." Say "Direct Stakeholders: Elderly shoppers (primary), Shop owners (secondary). Indirect Stakeholders: Family members helping them."
-
- **Question 2: "Draw an HTA for task X"**
  - **Action:** Draw the box diagram. Goal at top. Sub-tasks below.
  - **Trick:** Make sure you write the **PLAN** at the bottom (e.g., "Plan 0: Do 1, then 2..."). Without the Plan text, you lose marks.
-
- **Question 3: "Sketch the Interface"**
  - **Action:** Draw 3 squares.
    - Screen 1: Login/Home.
    - Screen 2: The list of groceries.
    - Screen 3: Payment success.
  -
  - **Trick:** Draw **Arrows** between the buttons on Screen 1 to the layout of Screen 2. Label them "User taps here." Examiners love arrows.
-
- **Question 4: "Which evaluation method should be used?"**
  - **Answer:** If the prototype is *sketchy*, say **Heuristic Evaluation** (cheap/fast). If the prototype is *finished*, say **Usability Testing** (accurate).
  - **Reason:** "Because we need to identify major usability issues before spending money on coding."
-
- **Question 5: "Justify the design based on Principles."**
  - **Answer:** Pick 3 words from the list: **Visibility** ("I made the font big so they can see"), **Feedback** ("It vibrates when clicked so they know it worked"), and **Constraint** ("I removed complex buttons to stop errors").
-

🚦 FINAL SURVIVAL TIPS (Read this on Feb 5 morning)

1. **Don't leave blanks.** If you forget the exact word for a Design Principle, describe it. (e.g., instead of "Affordance", say "The visual clue that suggests how to use it"). You will often still get the mark.
2. **Use specific vocabulary.** Use the word **"Iterative"** at least once. Use **"User-Centered"**. Use **"Cognitive Load"**.
3. **Read the graph:** If they give you a graph (like the Fitts' Law one or a Learning Curve), just explain what goes Up and what goes Down. "As the button gets smaller, time taken goes up." Simple.
4. **Prioritize the Past Papers:** The exact scenarios (e.g., Vending Machine, ATM, Booking System) often get recycled with only slight name changes.

Good luck with your other exams. On Feb 3rd at 5:00 PM, come back to this document. It is your lifeline.